

Anthropometric Study of Women During Menopause

A Descriptive Analysis

Cátia Sofia Fonseca Araújo, Escola de Engenharia da Universidade do Minho, ORCID: 0009-0004-1255-5932¹
Miguel Ângelo Fernandes Carvalho, Escola de Engenharia da Universidade do Minho, ORCID: 0000-0001-8010-6478²

Keywords: Woman; Menopause; Climacteric; Anthropometry; Sizing chart; Apparel industry.

Menopause is a natural stage of female ageing, characterized by the permanent cessation of ovarian function and reproductive capacity. The progressive decline in estrogen production triggers a range of physiological changes that affect multiple body systems, impacting not only women's overall health but also their body composition, morphology, and perception of body image.

Among the changes most commonly associated with the menopausal transition are increased body adiposity, redistribution of fat to the abdominal region, and alterations in body proportions. These changes may affect comfort, mobility, and garment fit, highlighting limitations in the sizing systems currently used by the textile and apparel industry.

In this context, a conceptual framework was developed to identify the key elements required for conducting anthropometric surveys focused on menopausal women, as well as the requirements necessary for designing a new sizing chart tailored to the characteristics and needs of this population and to the demands of the apparel industry.

Therefore, this poster aims to present the main factors involved in the anthropometric characterization of menopausal women, contributing to the development of more up-to-date and representative anthropometric data on their body dimensions. It is expected that this work will support the development of more appropriate sizing systems and promote inclusive, functional, ergonomic,

¹Cátia Sofia Fonseca Araújo, Escola de Engenharia da Universidade do Minho, ORCID: 0009-0004-1255-5932

² Miguel Ângelo Fernandes Carvalho, Escola de Engenharia da Universidade do Minho, ORCID: 0000-0001-8010-6478

and user-centered clothing design solutions that better meet the needs of women during this stage of life.

Referências

Costa, A. G. S., Canto Júnior, E. L., Costa Júnior, N. F., Sousa, R. J. A., Ribeiro, A. P. L., & Filgueiras, L. A. (2022). The influence of the Pilates method on postmenopausal women: literature review. *Health and Biosciences*, 3(1), 44–53. <https://doi.org/10.47456/hb.v3i1.37099>

HARLOW, S. D. et al. Executive summary of the Stages of Reproductive Aging Workshop +10: Addressing the unfinished agenda of staging reproductive aging. *Journal of Clinical Endocrinology & Metabolism*, v. 97, n. 4, p. 1159–1168, 2012. DOI: 10.1210/jc.2011-3362.

Lee BJ, Kim JU, Lee S. Association of menopausal status with body composition and anthropometric indices in Korean women. *PLoS One*. 2024 May 20;19(5):e0298212. doi: 10.1371/journal.pone.0298212. PMID: 38768131; PMCID: PMC11104629.

M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., ... Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, 372, n71. <https://doi.org/10.1136/bmj.n71>

Moilanen, J. M., Aalto, A. M., Hemminki, E., Aro, A. R., & Raitanen, J. (2010). Prevalence of menopause symptoms and their association with lifestyle among Finnish middle-aged women. *Maturitas*, 67(4), 368–374. <https://doi.org/10.1016/j.maturitas.2010.08.007>

Mulhall, S., Andel, R., Anstey, K. J., & Dixon, R. A. (2018). Transitions in depression and anxiety symptom trajectories: The role of menopausal status. *Menopause*, 25(9), 1021–1029. <https://doi.org/10.1097/GME.0000000000001105>

Nair, A., & colleagues. (2025). Patterns of bone mineral density loss at multiple skeletal sites following recent menopause in users and non-users of hormonal therapy. *Calcified Tissue International*.
<https://link.springer.com/article/10.1007/s00223-025-01392-8>

Schünemann, H. J., Tugwell, P., Reeves, B. C., Santesso, N. (2023). GRADE guidance 36: Updates to GRADE's approach to addressing inconsistency. *Journal of Clinical Epidemiology*, 158, 70-83.
<https://doi.org/10.1016/j.jclinepi.2023.03.003>

Wang K, Li Y, Chen Y. Anthropometric indices and mortality in postmenopausal women: NHANES 1999-2018 evidence. *BMC Womens Health*. 2025 Nov 26;25(1):623. doi: 10.1186/s12905-025-04111-5. PMID: 41299376; PMCID: PMC12751837.

Weber, M. T., Mapstone, M., Staskiewicz, J., & Maki, P. M. (2014). Reconciling subjective memory complaints with objective memory performance in the menopausal transition. *Menopause*, 21(9), 1-8.
<https://doi.org/10.1097/GME.0000000000000232>

Yan, D., Wang, X., & Wang, Y. (2024). The effectiveness of psychosocial interventions on non-physiological symptoms of menopause: A systematic review and meta-analysis. *Journal of Affective Disorders*, 363, 44-56.
<https://www.sciencedirect.com/science/article/pii/S0165032724003550>

Zhang, J., Wang, Y., Guo, J. et al. A associação entre dez medidas antropométricas e osteoporose e osteopenia em mulheres pós-menopáusicas. *Sci Rep* 15 , 10994 (2025). <https://doi.org/10.1038/s41598-025-94218-4>